

programmable switch (one P4 pipeline)

from master

Ingress

classify by
function code

match the
transaction

arm release
deadlines

forward the
request

read lane (READ, SELECT)

hold ACK, release at $t_A + D_A$

hold response, release + D_R

■■■■ blockers, strict priority

to master

on-chip packet generator

control lane (OPERATE)

hold OPERATE, to relay at $T_0 + J$

ACK at $T_0 + A$, echo at $T_0 + R$

■■■■ blockers, strict priority

to master

to relay at $T_0 + J$

unmatched: forward unchanged

forward

red: master-facing (observed) green: relay side orange: blockers from the generator